MSG No. 131.441 $P4_2/mm'c'$ [Type III, tetragonal]

Table 1: Wyckoff site: 2a, site symmetry: mm'm'.

No.	position	mapping
1	$[0,\ 0,\ 0]$	[1,4,5,8,9,10,13,14]
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	[2,3,6,7,11,12,15,16]

Table 2: Wyckoff site: 2b, site symmetry: mm'm'.

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[1,4,5,8,9,10,13,14]
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[2,3,6,7,11,12,15,16]

Table 3: Wyckoff site: 2c, site symmetry: mm'm'.

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	[1,4,5,8,9,10,13,14]
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[2,3,6,7,11,12,15,16]

Table 4: Wyckoff site: 2d, site symmetry: mm'm'.

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	[1,4,5,8,9,10,13,14]
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	[2,3,6,7,11,12,15,16]

Table 5: Wyckoff site: 2e, site symmetry: -4m'2'

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	[1,4,6,7,11,12,13,14]
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	[2,3,5,8,9,10,15,16]

Table 6: Wyckoff site: 2f, site symmetry: -4m'2'

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	[1,4,6,7,11,12,13,14]
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	[2,3,5,8,9,10,15,16]

Table 7: Wyckoff site: 4g, site symmetry: 2m'm'.

No.	position	mapping
1	$[0,\ 0,\ z]$	[1,4,13,14]
2	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	[2,3,15,16]

continued ...

Table 7

No.	position	mapping
3	$[0,\ 0,\ -z]$	`[5,8,9,10]`
4	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[6,7,11,12]`

Table 8: Wyckoff site: `4h`, site symmetry: `2m'm'`.

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,4,13,14]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[2,3,15,16]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[5,8,9,10]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[6,7,11,12]`

Table 9: Wyckoff site: `4i`, site symmetry: `2m'm'`.

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,4,13,14]`
2	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[2,3,15,16]`
3	$\left[0,\ \frac{1}{2},\ -z\right]$	`[5,8,9,10]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[6,7,11,12]`

Table 10: Wyckoff site: `4j`, site symmetry: `m2'm'`.

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,8,9,14]`
2	$\left[0,\ x,\ \frac{1}{2}\right]$	`[2,7,11,16]`
3	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[3,6,12,15]`
4	$[-x,\ 0,\ 0]$	`[4,5,10,13]`

Table 11: Wyckoff site: `4k`, site symmetry: `m2'm'`.

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[1,8,9,14]`
2	$\left[\frac{1}{2},\ x,\ 0\right]$	`[2,7,11,16]`
3	$\left[\frac{1}{2},\ -x,\ 0\right]$	`[3,6,12,15]`
4	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[4,5,10,13]`

Table 12: Wyckoff site: `4l`, site symmetry: `m2'm'`.

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[1,8,9,14]`
2	$[0,\ x,\ 0]$	`[2,7,11,16]`
3	$[0,\ -x,\ 0]$	`[3,6,12,15]`

continued ...

Table 12

No.	position	mapping
4	$\left[-x,\, 0,\, \frac{1}{2}\right]$	[4,5,10,13]

Table 13: Wyckoff site: 4m, site symmetry: m2'm'.

No.	position	mapping
1	$\left[x,\, \frac{1}{2},\, 0\right]$	[1,8,9,14]
2	$\left[\frac{1}{2},\, x,\, \frac{1}{2}\right]$	[2,7,11,16]
3	$\left[\frac{1}{2},\, -x,\, \frac{1}{2}\right]$	[3,6,12,15]
4	$\left[-x,\, \frac{1}{2},\, 0\right]$	[4,5,10,13]

Table 14: Wyckoff site: 8n, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\, x,\, \frac{1}{4}\right]$	[1,11]
2	$\left[-x,\, x,\, \frac{3}{4}\right]$	[2,10]
3	$\left[x,\, -x,\, \frac{3}{4}\right]$	[3,9]
4	$\left[-x,\, -x,\, \frac{1}{4}\right]$	[4,12]
5	$\left[-x,\, -x,\, \frac{3}{4}\right]$	[5,15]
6	$\left[x,\, -x,\, \frac{1}{4}\right]$	[6,14]
7	$\left[-x,\, x,\, \frac{1}{4}\right]$	[7,13]
8	$\left[x,\, x,\, \frac{3}{4}\right]$	[8,16]

Table 15: Wyckoff site: 8o, site symmetry: .m'.

No.	position	mapping
1	$[0,\, y,\, z]$	[1,13]
2	$\left[-y,\, 0,\, z+\frac{1}{2}\right]$	[2,15]
3	$\left[y,\, 0,\, z+\frac{1}{2}\right]$	[3,16]
4	$[0,\, -y,\, z]$	[4,14]
5	$[0,\, -y,\, -z]$	[5,9]
6	$\left[y,\, 0,\, \frac{1}{2}-z\right]$	[6,11]
7	$\left[-y,\, 0,\, \frac{1}{2}-z\right]$	[7,12]
8	$[0,\, y,\, -z]$	[8,10]

Table 16: Wyckoff site: 8p, site symmetry: .m'.

No.	position	mapping
1	$\left[\frac{1}{2},\, y,\, z\right]$	[1,13]
2	$\left[-y,\, \frac{1}{2},\, z+\frac{1}{2}\right]$	[2,15]
3	$\left[y,\, \frac{1}{2},\, z+\frac{1}{2}\right]$	[3,16]
4	$\left[\frac{1}{2},\, -y,\, z\right]$	[4,14]
5	$\left[\frac{1}{2},\, -y,\, -z\right]$	[5,9]

continued ...

Table 16

No.	position	mapping
6	$\left[y,\, \frac{1}{2},\, \frac{1}{2}-z\right]$	[6,11]
7	$\left[-y,\, \frac{1}{2},\, \frac{1}{2}-z\right]$	[7,12]
8	$\left[\frac{1}{2},\, y,\, -z\right]$	[8,10]

Table 17: Wyckoff site: 8q, site symmetry: m..

No.	position	mapping
1	$[x,\, y,\, 0]$	[1,8]
2	$\left[-y,\, x,\, \frac{1}{2}\right]$	[2,7]
3	$\left[y,\, -x,\, \frac{1}{2}\right]$	[3,6]
4	$[-x,\, -y,\, 0]$	[4,5]
5	$[x,\, -y,\, 0]$	[9,14]
6	$[-x,\, y,\, 0]$	[10,13]
7	$\left[y,\, x,\, \frac{1}{2}\right]$	[11,16]
8	$\left[-y,\, -x,\, \frac{1}{2}\right]$	[12,15]

Table 18: Wyckoff site: 16r, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$\left[-y,\, x,\, z+\frac{1}{2}\right]$	[2]
3	$\left[y,\, -x,\, z+\frac{1}{2}\right]$	[3]
4	$[-x,\, -y,\, z]$	[4]
5	$[-x,\, -y,\, -z]$	[5]
6	$\left[y,\, -x,\, \frac{1}{2}-z\right]$	[6]
7	$\left[-y,\, x,\, \frac{1}{2}-z\right]$	[7]
8	$[x,\, y,\, -z]$	[8]
9	$[x,\, -y,\, -z]$	[9]
10	$[-x,\, y,\, -z]$	[10]
11	$\left[y,\, x,\, \frac{1}{2}-z\right]$	[11]
12	$\left[-y,\, -x,\, \frac{1}{2}-z\right]$	[12]
13	$[-x,\, y,\, z]$	[13]
14	$[x,\, -y,\, z]$	[14]
15	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	[15]
16	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[16]